

Strategic Technology Cooperation in the U.S.-Japan-ROK Triangle

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Why It Matters

Critical and emerging technologies (CET) now shape the basic terms of national competition. They determine who can compute at scale, keep supply chains running under pressure, and turn scientific advantage into strategic influence. Among today's technology partnerships, the U.S.-Japan-South Korea triangle is especially well placed. The Harvard Belfer Center's 2025 Critical and Emerging Technologies Index highlights that no single country controls the full supply chain for advanced semiconductors, suggesting an important starting point where the three countries can face one another as practical partners.¹ This is clearest around Artificial Intelligence (AI). The United States leads in frontier AI models, chip design, and private capital. Japan brings strength in semiconductor equipment, materials, and precision manufacturing. South Korea is central to memory, especially the high bandwidth memory that modern AI accelerators rely on. AI centered cooperation therefore offers a promising first layer for a broader technology agenda, connecting models, chips, materials, infrastructure, and talent. Government strategies point in the same direction. Washington's updated CET List and AI Action Plan, Seoul's First Master Plan for twelve national strategic technologies, and Tokyo's Seventh Science, Technology and Innovation Basic Plan all identify overlapping priorities in AI, semiconductors, biotechnology, and quantum.² The 2023 Camp David Summit gave these shared interests an institutional starting point for trilateral cooperation in CETs.³

What the three countries want is broadly similar. Each seeks the high ground in the technologies that will define the next stage of economic and strategic power. The harder question is how to make cooperation work when each country is also running its own national race. That requires a clear understanding of comparative advantage and cooperation channels that extend beyond government-to-government ties. More attention should fall on the actual builders of frontier technology, from major firms and start-ups to research

¹ Eric Rosenbach et al., "Critical and Emerging Technologies Index" (Belfer Center for Science and International Affairs, Harvard Kennedy School, June 5, 2025), <https://www.belfercenter.org/critical-emerging-tech-index>.

² Fast Track Action Subcommittee on Critical and Emerging Technologies, Critical and Emerging Technologies List Update (Washington, DC: National Science and Technology Council, Executive Office of the President, February 2024), <https://bidenwhitehouse.archives.gov/wp-content/uploads/2024/02/Critical-and-Emerging-Technologies-List-2024-Update.pdf>; Office of Science and Technology Policy, Winning the Race: America's AI Action Plan (Washington, DC: The White House, July 2025), <https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf>; Ministry of Science and ICT, "MSIT Unveils First Master Plan for Developing Critical and Emerging Technologies (2024–2028): A Blueprint for National S&T Sovereignty," press release, August 26, 2024, <https://www.msit.go.kr/eng/bbs/view.do?sCode=eng&mId=4&bbsSeqNo=42&nttSeqNo=1034>; Cabinet Office, Government of Japan, Seventh Science, Technology and Innovation Basic Plan, cabinet decision, March 27, 2026, <https://www8.cao.go.jp/cstp/kihonkeikaku/7honbun.pdf>.

³ U.S. Embassy and Consulate in the Republic of Korea, "The Spirit of Camp David: Joint Statement of Japan, the Republic of Korea, and the United States," August 19, 2023, <https://kr.usembassy.gov/081923-the-spirit-of-camp-david-joint-statement-of-japan-the-republic-of-korea-and-the-united-states/>.

campuses and specialized labs, where new systems are designed, tested, scaled, and commercialized. In this setting, governments remain essential, but their role is less to direct innovation than to make it possible. Cutting-edge and resource-heavy, these fields depend on governments to set basic guidelines, ease regulatory barriers where needed, and provide funding, while allowing firms, universities, and laboratories to carry the work forward.

AI and Semiconductors: AI Infrastructure as the First Test

AI infrastructure is the most immediate arena for trilateral cooperation, given how deeply AI now connects to other frontier fields. The key bottleneck increasingly lies not only in software and algorithms, but in accelerators, memory chips, data centers, energy supply, and the operating layer that keeps models running at scale. This is where the three economies find their most natural point of convergence. Much of this cooperation is already being carried out by the actors that hold the technology itself, including firms, research institutes, and universities. NVIDIA's multiyear memory partnerships with SK Hynix and Samsung show this interdependence in commercial practice, while firm and university ties extend it into adjacent sectors.⁴ Hyundai's ownership of Boston Dynamics and Toyota's partnership with Waymo connect the three countries in robotics and mobility,⁵ while academic networks such as the SNU-CMU Human Centered AI Research Center and U.S.-Japan university-corporate partnerships involving Carnegie Mellon, Keio, the University of Washington, and Tsukuba help root cooperation in the institutions that train future talent.⁶ In this setting, governments are less the drivers of cooperation than its backstop. Their role is to help cooperation start, make it easier to sustain, and ensure that funding and rules do not become obstacles. Governments do not need to manage every exchange directly, but they do need to keep the channels open. Yet the current shape of cooperation also reveals an imbalance. Many of today's links branch out from the United States as the hub, with South Korea and Japan each connected to it more strongly than to each other. This pattern is understandable, given the United States' lead in AI-related areas, but it also leaves the South Korea-Japan side of the triangle comparatively thin. Strengthening that link would make the partnership more balanced, durable, and capable of operating beyond U.S.-centered channels.

Biotechnology: Securing the Chain After Discovery

⁴ NVIDIA, "NVIDIA and SK hynix Announce Multiyear Technology Partnership to Advance Memory for AI Factories," NVIDIA Newsroom, June 8, 2026, <https://nvidianews.nvidia.com/news/sk-hynix-ai-factory>; NVIDIA, "NVIDIA and Samsung Build AI Factory to Transform Global Intelligent Manufacturing," NVIDIA Newsroom, 2025, <https://nvidianews.nvidia.com/news/samsung-ai-factory>.

⁵ Hyundai Motor Group, "Hyundai Motor Group Completes Acquisition of Boston Dynamics From SoftBank," newsroom, June 21, 2021, <https://www.hyundai.com/worldwide/en/newsroom/detail/hyundai-motor-group-completes-acquisition-of-boston-dynamics-from-softbank-0000000516>; Waymo, "Waymo and Toyota Outline Strategic Partnership," Waymo Blog, April 2025, <https://waymo.com/blog/2025/04/waymo-and-toyota-outline-strategic-partnership/>.

⁶ Carnegie Mellon University, "CMU Launches Human-Centered AI Research Center With Seoul National University," Human-Computer Interaction Institute, February 2025, <https://hcii.cmu.edu/news/cmu-snu-human-centered-ai-research-center>; U.S. Embassy Tokyo, "United States and Japan Announce Two New University-Corporate AI Partnerships Worth \$110 Million," April 9, 2024, <https://jp.usembassy.gov/us-japan-announce-2-university-corporate-ai-partnerships-worth-110-million/>.

In biotechnology, the strategic value of trilateral cooperation lies less in discovery alone than in what follows it. Tools such as Google DeepMind's AlphaFold have turned protein-structure prediction into a more open and shared foundation, one that researchers in South Korea and Japan are already using to predict proteins and identify previously unknown viruses from environmental data.⁷ The next challenge is to turn these scientific gains into products that can be developed, approved, manufactured, and delivered across borders. That process is rarely linear, as the chain faces obstacles at every stage, from specialized production facilities and clinical standards to confidential data and government regulations. These frictions are compounded by growing competition among major technology and pharmaceutical blocs, making coordination among trusted partners central to the field's future. Recent moves by the United States, Japan, and South Korea show that this agenda is already moving onto firmer ground. Industrial partnerships such as Moderna's work with Samsung Biologics and Novavax's cooperation with Takeda on COVID-19 vaccines show how cooperation can operate at the production stage.⁸ The 2024 Biopharmaceutical Alliance and the biotechnology provisions of the 2025 U.S.-Japan and U.S.-ROK Technology Prosperity Deals push the agenda further toward supply-chain resilience and trusted coordination.⁹ Korean ventures entering Japan's Shonan iPark carry the same logic into the start-up and research ecosystem.¹⁰ Given that pharmaceutical supply chains can bear directly on human lives, governments should focus first on strengthening coordination among health authorities and managing regulatory barriers before they slow the chain down. At the same time, as three countries with deep strengths in biomedical research, they should continue funding fundamental research and encouraging joint projects that also keep the discovery pipeline moving.

Quantum: Building the Ecosystem Before the Market

Quantum technologies are more anticipatory than most frontier technologies. They are not yet transforming industries the way AI has, but governments are already preparing for the possibility that their long-term impact could be even larger. This unique gap between future promise and present commercialization shapes the field's structure. With no clear commercial payoff yet, private firms invest cautiously, while much of

⁷ John Jumper et al., "Highly Accurate Protein Structure Prediction with AlphaFold," *Nature* 596, no. 7873 (2021): 583–589, <https://doi.org/10.1038/s41586-021-03819-2>; Google, "Researchers in Asia Pacific Use AlphaFold to Advance Scientific Discovery," *The Keyword*, June 2025, <https://blog.google/company-news/inside-google/around-the-globe/google-asia/researchers-asia-pacific-alphafold-five-years/>.

⁸ Samsung Biologics, "Samsung Biologics Signs Agreement with Moderna for Fill-Finish Manufacturing of COVID-19 Vaccine," company news, 2021, <https://samsungbiologics.com/kr/media/company-news-view?boardseq=1207>; Takeda Pharmaceutical Company, "Novavax and Takeda Announce Collaboration for Novavax COVID-19 Vaccine Candidate in Japan," news release, 2020, <https://www.takeda.com/newsroom/newsreleases/2020/novavax-and-takeda-announce-collaboration-for-novavax-covid-19-vaccine-candidate-in-japan/>.

⁹ "South Korea, US, Japan, India and EU Launch Biopharmaceutical Alliance," *Science|Business*, 2024, <https://sciencebusiness.net/network-updates/south-korea-us-japan-india-eu-launch-biopharmaceutical-alliance>; The White House, "U.S.-Japan Technology Prosperity Deal," October 2025, <https://www.whitehouse.gov/releases/2025/10/u-s-japan-technology-prosperity-deal/>; The White House, "U.S.-Korea Technology Prosperity Deal," October 2025, <https://www.whitehouse.gov/releases/2025/10/u-s-korea-technology-prosperity-deal/>.

¹⁰ iPark Institute Co., Ltd., "Eight Cutting-edge South Korean Bio Startups Begin Moving into Shonan iPark," press release, September 27, 2024, <https://www.shonan-ipark.com/en/news/2063/>.

the momentum still comes from government funding channeled through universities and the wider research community. American firms, most visibly IBM, are sharing advanced quantum systems with researchers worldwide, in part to make their own platforms the field's working standard. Building on Japan's strength in high-performance computing, RIKEN now operates the first IBM Quantum System Two outside the United States, co-located with the Fugaku supercomputer to pioneer hybrid quantum-classical computing, while Yonsei University hosts Korea's first on-premises quantum platform for research and talent development.¹¹ The 2023 U.S.-ROK quantum joint statement, the U.S.-Japan Tokyo Statement on Quantum Cooperation, and the recent Technology Prosperity Deals all show that the three governments have already reached a broad common understanding of quantum technology's future strategic importance.¹² At this still-early stage, trilateral quantum cooperation should remain centered on academia and leading firms by training the next generation of researchers, expanding shared access to experimental platforms, and building the human foundations of a future quantum ecosystem. At the same time, the three governments should begin preparing for the regulatory challenges and long-term risks that quantum technologies may eventually create.

Looking Forward: Four Commitments for Durable Cooperation

The institutional foundation built at Camp David and extended through subsequent bilateral agreements is already in place. What matters now is whether this architecture can move beyond summit declarations and become a habit of cooperation. Four commitments should guide that shift.

First, governments should set the field, not play the match themselves. Frontier technologies are built by firms, universities, laboratories, investors, and start-ups, each bringing capacities that governments cannot replace. The role of the state is to provide rules, resources, and room to move. Across the United States, Japan, and South Korea, that means clarifying research-security guidelines, lowering barriers to cross-border investment, sustaining public funding, supporting shared infrastructure, and ensuring that all three sides of the triangle grow thicker, not only the links running through Washington.

Second, the three countries should treat cross-border talent as a long-term strategic asset. Bringing together researchers and engineers trained in different systems would allow each country's strengths to be absorbed, tested, and recombined in ways that stay difficult within national programs alone. Talent exchange should not be viewed only through the lens of brain drain or technology leakage. It should be understood as a way

¹¹ IBM, "IBM and RIKEN Unveil First IBM Quantum System Two Outside of the U.S.," IBM Newsroom, June 23, 2025, <https://newsroom.ibm.com/2025-06-23-ibm-and-riken-unveil-first-ibm-quantum-system-two-outside-of-the-u-s>; IBM, "Yonsei Deploys First IBM Quantum System One in the Republic of Korea," IBM Newsroom, November 19, 2024, <https://newsroom.ibm.com/2024-11-19-yonsei-deploys-first-ibm-quantum-system-one-in-the-republic-of-korea>.

¹² U.S. Department of State, "Joint Statement of the United States of America and Republic of Korea on Cooperation in Quantum Information Science and Technologies," April 26, 2023, <https://2021-2025.state.gov/joint-statement-of-the-united-states-of-america-and-republic-of-korea-on-cooperation-in-quantum-information-science-and-technologies/>; U.S. Department of State, "Tokyo Statement on Quantum Cooperation," 2023, <https://2021-2025.state.gov/tokyo-statement-on-quantum-cooperation/>; The White House, "The United States Signs Technology Prosperity Deals with Japan and Korea," October 2025, <https://www.whitehouse.gov/releases/2025/10/the-united-states-signs-technology-prosperity-deals-with-japan-and-korea/>.

for young researchers to experience each country's strengths firsthand, build durable professional ties, and carry those networks into the next generation of technological cooperation.

Third, governance must move at the speed of the technologies it seeks to guide. In frontier sectors, a single quarter can be enough for a field to enter a new phase. Policymakers should therefore work more closely with technical experts and cultivate interdisciplinary talent that combines hands-on research experience with responsibility for the social, economic, and security consequences of new technologies. Rather than trying to stop or control the current of technological change, the three governments should learn to steer it together.

Finally, cooperation must outlast any single government. Science and technology diplomacy, industrial coordination, and university research operate on timelines longer than electoral cycles. The three countries should avoid treating joint programs as short-term political announcements. Once set in motion, cooperation should be sustained through institutions, funding channels, and recurring exchanges that can endure beyond changes in administration.

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